
M⁴oE: A Foundation Model for Medical Multimodal Image Segmentation with Mixture of Experts

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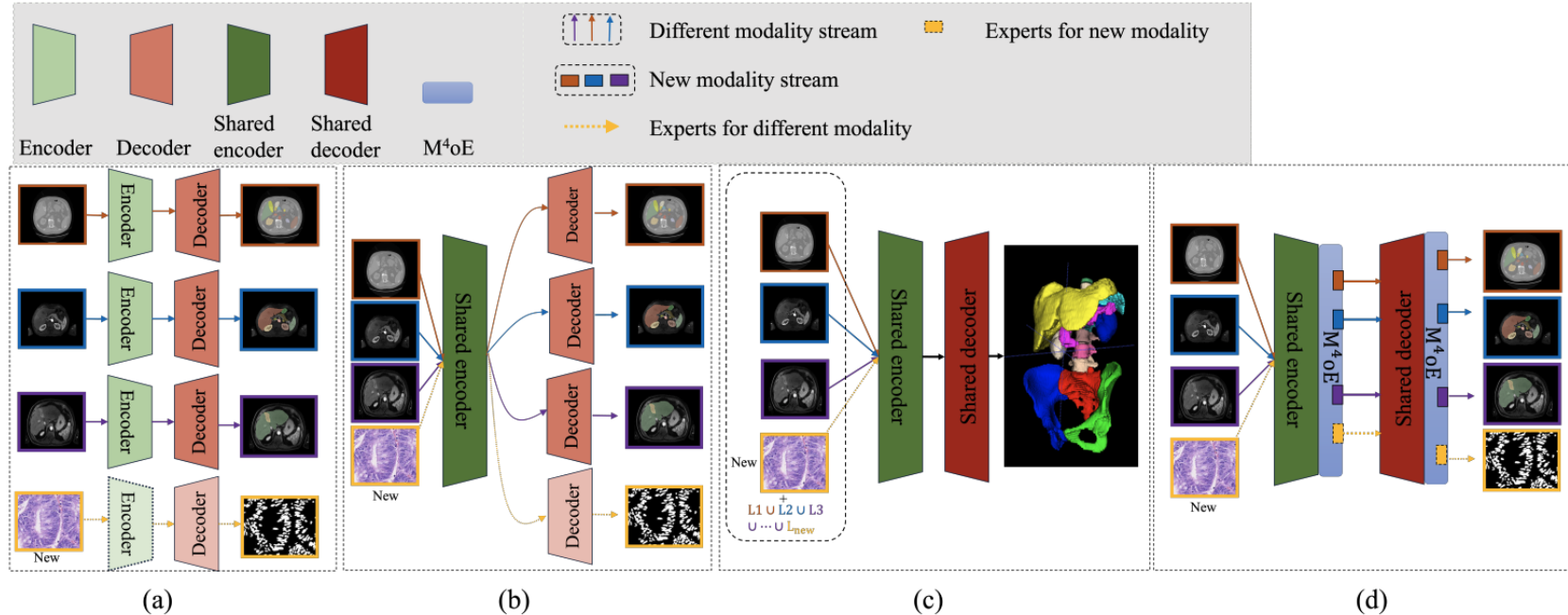
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Abstract

Medical imaging data is inherently heterogeneous across different modalities and clinical centers, posing unique challenges for developing generalizable foundation models. Conventional entail training distinct models per dataset or using a shared encoder with modality-specific decoders. However, these approaches incur heavy computational overheads and suffer from poor scalability. To address these limitations, we propose the Medical Multi-Modal Mixture of Experts (M4oE) framework, leveraging the SwinUNet architecture. Specifically, M4oE comprises modality-specific experts, each separately initialized to learn features encoding domain knowledge. Subsequently, a gating network is integrated during fine-tuning to dynamically modulate each expert's contribution to the collective predictions. This enhances model interpretability as well as the generalization ability while retaining expertise specialization. Simultaneously, the M4oE architecture amplifies the model's parallel processing capabilities, and it also ensures the model's adaptation to new modalities with ease. Experiments across three modalities reveal that M4oE can achieve 3.45% over STU-Net-L, 5.11% over MED3D, and 11.93% over SAM-Med2D across the MICCAI FLARE22, AMOS2022, and ATLAS2023 datasets. Moreover, M4oE showcases a significant reduction in training duration with 7 hours less, while maintaining a parameter count that is only 30% of its compared methods. The code is available at <https://github.com/JefferyJiang-YF/M4oE>.

- Introduction
- Methods
- Results
- Conclusion
- Limitation

- Vision Foundation Models:
 - 자연 이미지 이해는 잘하지만 그 이해를 의료 영상에 전이하는 것은 어려움:
 - The heterogeneity across modalities like MRI, CT, and X-rays (데이터의 특성이 서로 크게 다름)
 - 의료 영상은 자연 이미지용으로 설계된 모델의 능력을 넘어서는 특수한 처리가 필요함.
- 일반 Transformer는 modality-specific attributes를 최적으로 adopt하지 못함.
 - 이를 Mixture of Experts(MoE)를 사용하여 각 modality의 특징에 집중하도록 설계하여 large heterogeneous datasets으로 효율적으로 스케일링하도록 함.



방식	장점	단점
(a) 모달리티별 독립 네트워크	초기 복잡도 낮음	새 데이터셋 적용 시 오버헤드 큼
(b) 공유 인코더 + 모달리티별 디코더	공통 표현 추출 맞춤 처리	데이터 다양성 증가 시 연산량 비례하여 증가
(c) 공유 인코더 + 공유 디코더	레이블 처리 표준화	새 모달리티 확장성 저하
M4oE 공유 인코더 + 공유 디코더 + MoE	병렬 처리 ↑ 새 모달리티 추가 용이	

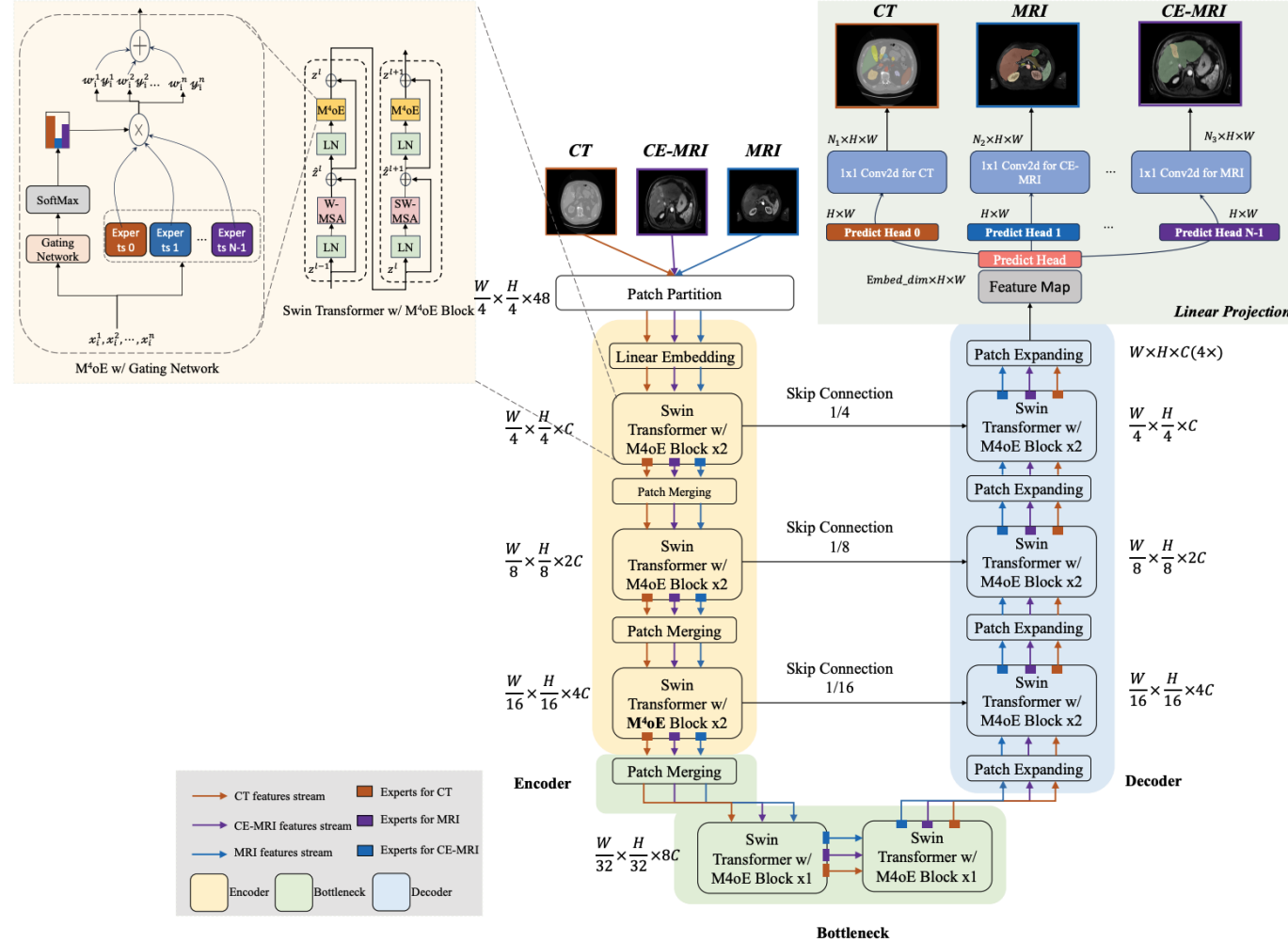
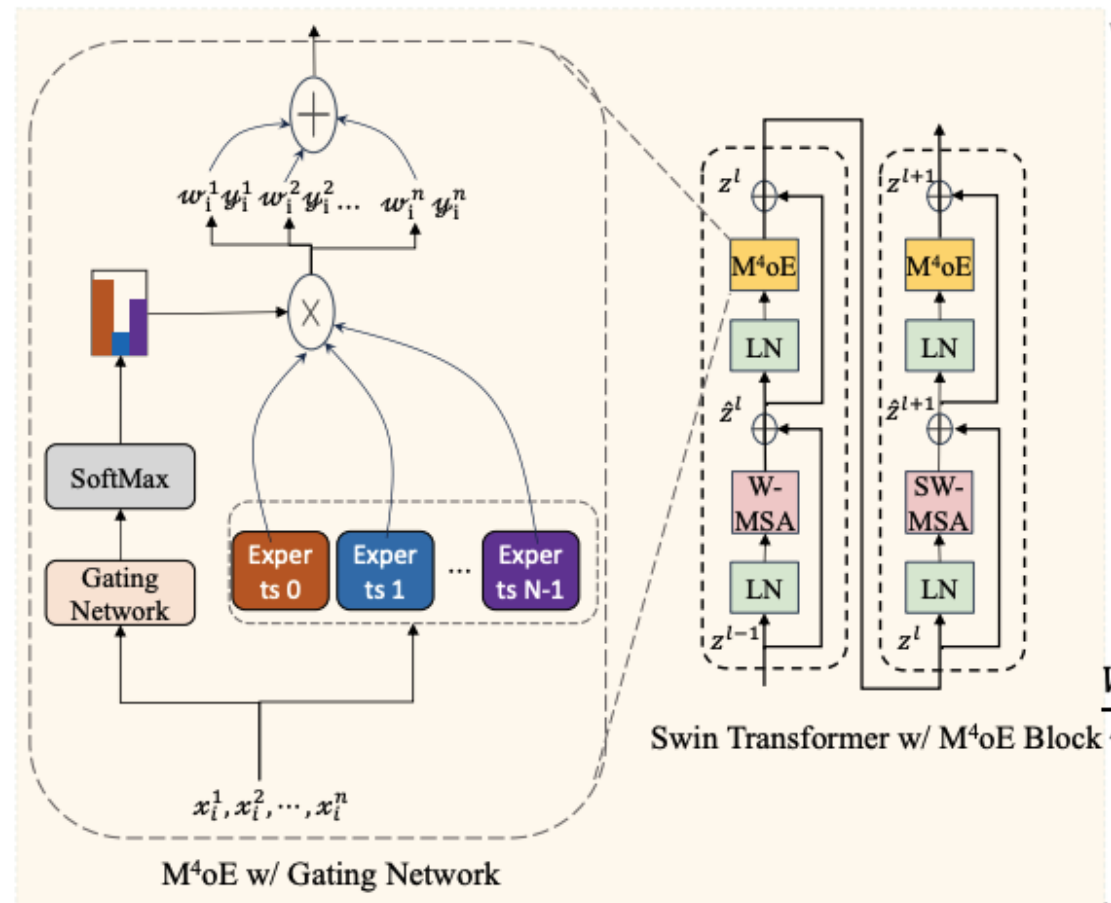


Fig. 2. Overall framework of M⁴oE implemented on a SwinUNet backbone. At the core is the M⁴oE with a gating network, which dynamically selects specialized experts for different imaging modalities (CT, CE-MRI, MRI); each expert learns representation for its respective data characteristics.

- Swin Transformer Backbone
 - 일반적인 multi-headed selfattention (MHSA)와 달리 Swin Transformers는 window-based MHSA (W-MSA)라는 로컬 슬라이딩 윈도우 방식을 사용함.



- Medical Multi-Modal Mixture-of-Experts (M⁴oE)

```

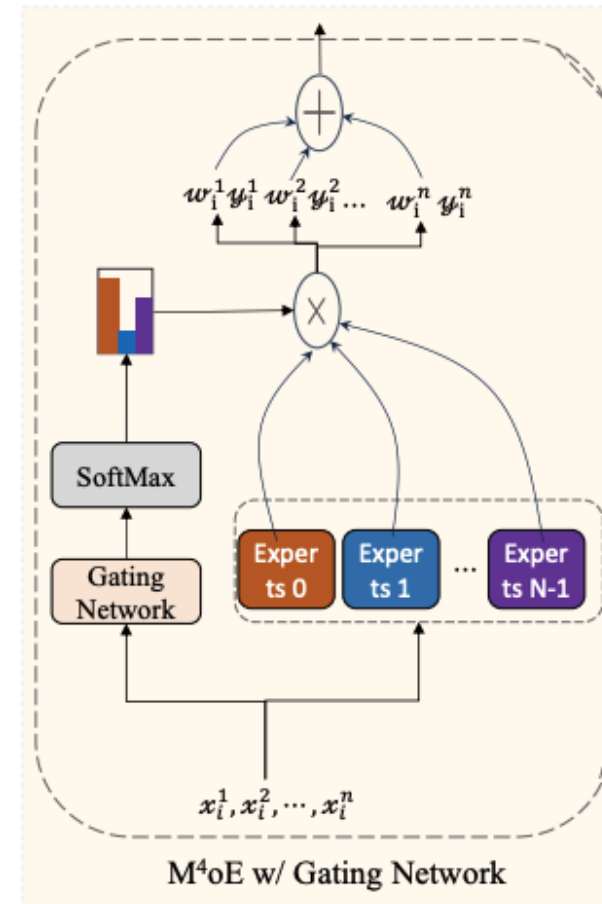
# MoE with gating network
class MoEFFN_Gating(nn.Module):
    def __init__(self, dim, hidden_dim, num_experts):
        super(MoEFFN_Gating, self).__init__()
        self.gating_network = nn.Linear(dim, dim)
        self.experts = nn.ModuleList([nn.Sequential(
            nn.Linear(dim, hidden_dim),
            nn.GELU(),
            nn.Linear(hidden_dim, dim)
        ) for _ in range(num_experts)])

    def forward(self, x, dataset_id):
        # Gating mechanism to determine the mixture weights
        weights = self.gating_network(x)
        weights = torch.nn.functional.softmax(weights, dim=-1)

        # Log the weights
        # self.log_weights(weights)

        # Get outputs from all experts
        outputs = [expert(x) for expert in self.experts]
        outputs = torch.stack(outputs, dim=0)

        # combine the experts' outputs
        # print(weights)
        outputs = (weights.unsqueeze(0) * outputs).sum(dim=0)
        return outputs
  
```



- Dynamic Selection for Linear Projection Head
 - Multimodal segmentation의 core challenge는 modalities간 클래스 정보가 다양한 것임.
 - Feature map padding: 하여 모든 샘플의 차원을 동일하게 맞춤
 - Dynamic selection: 각 샘플의 linear projection이 해당 모달리티에만 필요한 클래스 logit만 선택
 - 출력 차원이 원래 레이블 분포에 정렬됨

```

def up_x4(self, x, predict_head):
    H, W = self.patches_resolution
    B, L, C = x.shape
    assert L == H * W, "input features has wrong size"

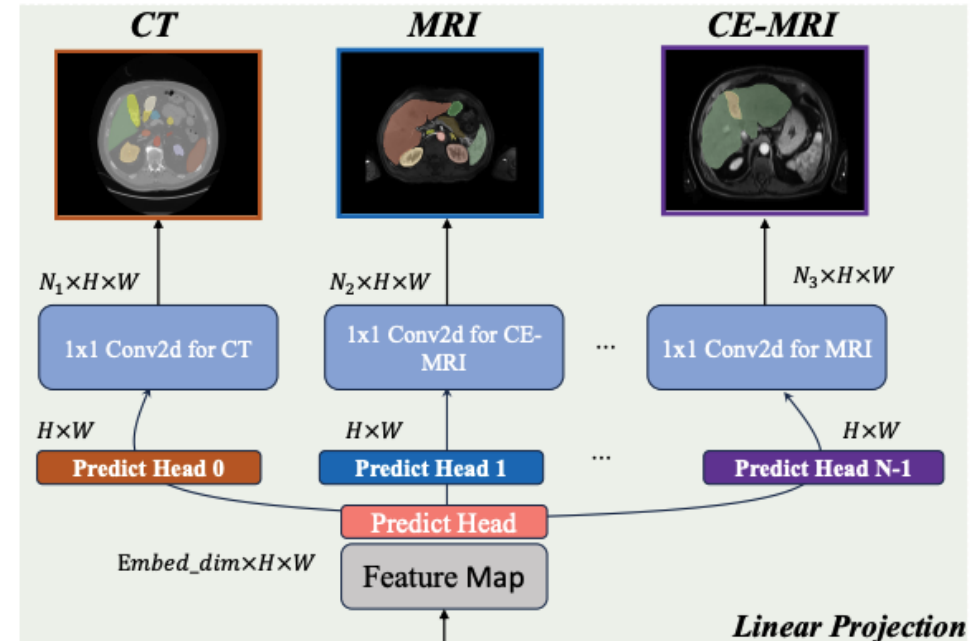
    if self.final_upsample == "expand_first":
        x = self.up(x)
        x = x.view(B, 4 * H, 4 * W, -1)
        x = x.permute(0, 3, 1, 2) # B,C,H,W

    output = []
    max_channels = 0
    for i in range(B):
        output_layer = self.outputs[str(predict_head[i].item())]
        output_i = output_layer(x[i].unsqueeze(0))
        output.append(output_i)
        if output_i.size(1) > max_channels:
            max_channels = output_i.size(1)

    # Pad the outputs to the same number of channels
    for i in range(B):
        padding = max_channels - output[i].size(1)
        output[i] = F.pad(output[i], (0, 0, 0, 0, 0, padding))

    # print("output_i.shape:", output[i].shape)
    x = torch.cat(output, dim=0)

    return x
  
```



```

# self.output = nn.Conv2d(in_channels=embed_dim, out_channels=out_channels, kernel_size=1, bias=False)
self.outputs = nn.ModuleDict({
    str(i): nn.Conv2d(in_channels=embed_dim, out_channels=num_classes[i], kernel_size=1, bias=False)
    for i in range(len(num_classes))
})
  
```

- Env
 - GPU: RTX 4090(CUDA 11.3)
 - PyTorch 1.11.0
 - Python 3.8
 - Ubuntu 20.04.
- Pre-training
 - Encoder: MAE
- Train
 - Decoder: SwinUNet 원본과 동일
 - Batch size: 36
 - Optimizer: AdamW
 - weight decay: $1e-4$
- Detail
 - $\text{loss} = 0.4 * \text{loss}_{\text{ce}} + 0.7 * \text{loss}_{\text{dice}}$

```
model = model.to(device)
optimizer = optim.SGD(model.parameters(), lr=base_lr, momentum=0.9, weight_decay=0.0001)
writer = SummaryWriter(snapshot_path + '/log')
```

- Datasets
 - 3 multimodal medical image segmentation datasets:
 - FLARE22
 - 50 labeled CT scan cases
 - AMOS22
 - 600 labeled cases comprising 500 CT and 100 MRI abdominal scans
 - ATLAS23
 - 60 cases of contrast-enhanced MRI (CEMRI) liver scans
 - data split: previous work
- FLARE22와 AMOS22는 복부인 반면, ATLAS23은 간에 특화됨.
 - 이를 통해 M⁴oE의 다양한 anatomical segmentation tasks를 처리하는 능력을 확인 가능

- Results

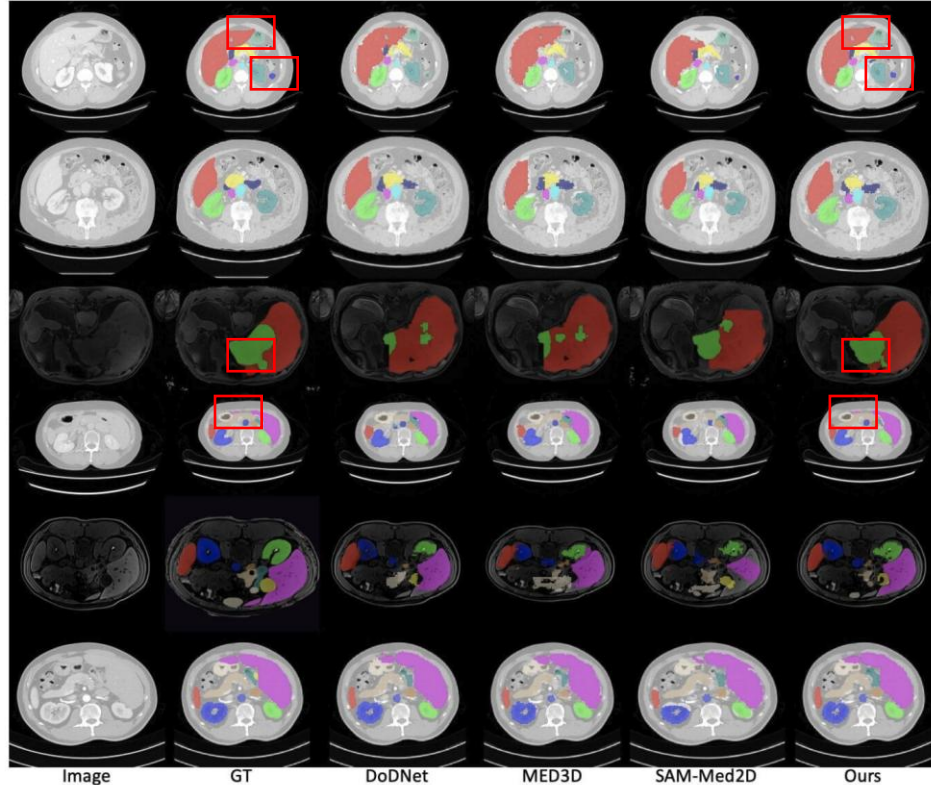


Fig. 3. Visualization of segmentation results from various methods. The first and second rows depict FLARE22 with CT images, the third row showcases ATLAS2023 with CE-MRI images, the fourth and sixth rows display AMOS2022 with CT images, and the fifth row presents MRI images from AMOS2022. The six columns from left to right correspond to the original image, the ground truth (GT), the DoDNet results, the MED3D results, the SAM-MED2D results, and our M⁴oE results.

Table 1. Comparison of M⁴oE against prior arts in terms of crucial model design choices and computational requirements, including pre-training strategy, input dimensionality, multimodality handling capability, number of parameters, training duration, and parameters added when introducing new modalities.

Methods	Pre-train	2D/3D	Multimodality	#Param	Training hours	Introduce new modality
DoDNet [20]	×	3D	✓	17.3M	48 hours	-
MED3D [3]	×	3D	✓	117.51M	16 hours	21M
Liu's [13]	×	3D	×	6.66M	19.5 hours	6.66M
STU-Net-L [9]	✓ Supervised	3D	×	440.30M	156.8 hours	440.30M
Ours	✓ Self-supervised	2D	✓	27.31M	12 hours	10M

Table 2. Comparison of DSC (%) across multimodal segmentation methods. Our M⁴oE model outperforms others, achieving the highest DSC in FALRE and AMOS-CT tasks. It shows adaptability and improvement (*) upon introducing a new modality, contrary to DoDNet and MED3D, which see performance drops. Unlike STU-Net-L's single-dataset focus, our model's robustness is attributed to training on three varied datasets. SAM-Med2D's performance is evaluated using a single data point. Results are highlighted in red for best and blue for second best.

Datasets	Method	Liver	Left Kidney	Right Kidney	Gallbladder	Pancreas	Spleen	Stomach	Mean
FLARE22	DoDNet [20]	96.64	95.43	95.32	74.58	79.48	90.91	90.85	89.03
	MED3D [3]	90.36	87.23	88.47	67.84	73.82	91.98	92.30	84.53
	Liu's [13]	89.30	76.32	78.40	55.35	58.86	83.07	67.82	72.73
	STU-Net-L [9]	94.48	87.51	86.68	76.31	82.00	91.27	89.31	86.79
	SAM-Med2D [4]	80.37	77.96	78.19	50.37	53.12	82.47	75.01	71.07
	Ours	97.08	94.62	94.92	79.38	76.99	95.74	89.78	89.78
AMOS-CT	DoDNet [20]	92.14	83.93	82.48	70.73	73.69	90.58	85.14	82.67
	MED3D [3]	88.25	85.11	85.29	65.12	59.78	88.37	81.10	79.00
	STU-Net-L [9]	95.47	88.37	89.74	76.92	82.64	93.24	87.03	87.63
	SAM-Med2D [4]	67.51	71.09	68.31	57.93	51.24	76.12	72.93	66.45
	Ours	95.59	88.72	88.31	73.94	61.39	90.59	82.78	83.04
AMOS-CT+MRI*	DoDNet [20]	89.31	80.33	82.89	70.13	68.32	85.80	83.23	80.01
	MED3D [3]	83.42	82.11	86.33	69.12	63.20	86.23	80.79	78.74
	STU-Net-L [9]	95.59	89.00	90.10	79.97	94.08	94.37	94.42	91.08
	SAM-Med2D [4]	55.29	63.27	62.57	48.18	50.12	61.29	68.10	58.40
	Ours	94.38	89.20	88.16	76.37	69.12	91.23	85.34	84.83

- Results

Table 3. Results on different methods on ATLAS2023, we use one point to test SAM-Med2D on test sets. The DoDNet and MED3D are not robust when the new modality is introduced. The text is in **red** and in **blue** for the best and second-best results.

Methods	DSC (%)			mIoU (%)		
	Liver	Tumor	Mean	Liver	Tumor	Mean
DoDNet[20]	79.54	24.03	52.34	75.31	19.3	47.31
MED3D [3]	73.12	19.55	46.34	65.3	16.82	41.06
SAM-Med2D [4]	80.02	51.28	65.65	77.34	48.88	63.11
Ours	89.43	57.53	73.48	86.34	55.32	70.83

Table 4. Ablation study on different model configurations. Encoder or decoder marked with **✗** indicates the use of an MLP as the output layer for the Swin Transformer block. In contrast, **✓** denotes that the encoder or decoder incorporates our M⁴oE structure. 'GN' stands for gating network. The ablation studies reveal that employing the M⁴oE architecture in both encoder and decoder significantly enhances model performance. The text is in **red** and in **blue** for the best and second-best results.

Method				DSC (%)								
Encoder	Decoder	GN	M ⁴ oE	Liver	Liver Tumor	Left Kidney	Right Kidney	Gallbladder	Pancreas	Spleen	Stomach	Mean
✗	✗	✗	✗	93.61	51.23	84.65	84.51	69.34	63.21	92.37	81.20	77.52
✓	✗	✓	✓	93.21	55.12	89.27	89.15	75.41	68.32	91.12	83.25	80.61
✓	✗	✗	✓	93.58	53.21	90.01	90.60	76.01	65.33	90.87	82.79	80.30
✗	✓	✓	✓	94.08	53.25	88.98	89.01	75.76	69.31	92.37	85.81	81.07
✗	✓	✗	✓	94.30	53.19	90.12	88.91	74.11	68.45	91.36	85.43	80.73
✓	✓	✗	✓	93.12	55.37	88.67	87.56	75.39	66.29	90.12	83.04	79.95
✓	✓	✓	✓	94.12	57.53	90.85	90.46	76.56	69.16	92.52	85.97	82.15

- M⁴oE는 multimodal medical image segmentation foundation models 개발에 있어 새롭고 효율적인 방향을 제시함.
- 유연한 게이팅 메커니즘 내에서 모달리티별 전문가를 효과적으로 통합함으로써, M⁴oE는 다양한 영상 데이터셋에 걸쳐 정확하고 적응적인 분할을 보여줌.
- 여기서 제시된 expert specialization and dynamic ensemble 개념이 의료 영상 및 그 이상의 다중 모달 과제를 해결하기 위한 지침 원칙을 제공할 것으로 기대함.
- 향후 방향:
 - 대규모 데이터셋에서 M⁴oE 평가
 - uncertainty estimation
 - enhancing model transparency

- 3D가 아닌 2D 기반 framework
- MAE pretraining에 대한 설명 부족
 - 어떤 데이터로, 어떤 방식으로 각 expert를 분리 학습했는지 불명확함.
- Data preprocessing, augmentation에 대한 설명 부족
 - Issue에 무수한 요청이 있었으나 ...